# A second-order and moment proof of the five-point Mordell inequality

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**Abstract**

We give a computer-assisted proof of the sharp five-point Mordell discriminant inequality by using second-order conditions and low-degree moments at a maximizer. A general variational estimate bounds the variance of the squared radii and the second complex power sum. For five points, a two-dimensional projection supplies a constraint connecting the third and fourth power sums. A quartic majorant for the logarithm then reduces the global comparison to a two-variable inequality. Near the unit circle, an angular concavity estimate and a weighted Laplacian give a separate exclusion argument whose constants are verified algebraically. One scalar branch is excluded by a uniform rational estimate; the remaining branch is certified by exact rational interval arithmetic in a short Python program using only the standard library. The argument is independent of the classification of stationary configurations. We also give a moment-matrix hierarchy for arbitrary point counts and an explicit local radial gap for six points.

**Keywords:** discriminant inequality; Vandermonde product; second-order optimality; moment bounds; exact interval arithmetic.
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## 1 Introduction

For $n \geq 2$ and $z = (z_1, \ldots, z_n) \in \mathbb{C}^n$, write

$$\Delta_n(z) = \prod_{i<j} |z_i - z_j|^2, \qquad \mathcal{S}_n = \left\{ z : \sum_i |z_i|^2 = n \right\}.$$

The problem considered by Mordell [1] asks when the regular $n$-gon on the unit circle maximizes $\Delta_n$ on $\mathcal{S}_n$. For a monic polynomial with roots $z_i$, this product is the absolute value of its discriminant.

Wang [3] gives a proof of the five-point inequality by exact algebraic classification of the stationary equations, followed by discriminant comparison. The present paper gives a different proof of the same sharp inequality. Its starting point is the second-order condition at a maximizer. The analytic reduction uses a complex symmetric matrix, a two-dimensional projection, and a quartic logarithm majorant; the final certificate involves only two real variables.

**Theorem 1.1** (Five-point inequality; cf. [3])**.** *If $z_1, \ldots, z_5 \in \mathbb{C}$ and $\sum_i |z_i|^2 = 5$, then*

$$\Delta_5(z) \leq 5^5. \tag{1.1}$$

*Equality holds precisely when, after a permutation, $z_j = e^{i\theta} e^{2\pi i(j-1)/5}$ for some $\theta \in \mathbb{R}$.*

The proof is computer-assisted. Every reduction from (1.1) to the scalar inequalities is given below. The ancillary program `verify_mordell.py` certifies the constants and the final scalar inequality with rational arithmetic and integer square roots. It requires no computer algebra system or numerical optimizer. A separate ancillary script checks the distance-moment expansions and the polynomial encoding.

The general second-order estimates admit a hierarchy of higher-moment constraints, developed in Section 8. The local angular argument also extends to six points. These extensions concern the structure of local maximizers; the bound $\Delta_n \le n^n$ itself fails for $n \ge 6$, as shown by Chalk [2].

## 2 Second-order estimates

### 2.1 A general radial bound

We use the Hermitian inner product $\langle x, y\rangle = \sum_i \overline{x_i} y_i$; a superscript $\mathsf{T}$ denotes transpose, and $*$ denotes conjugate transpose. Powers of vectors below are taken coordinatewise.

**Proposition 2.1.** *Let $n \ge 2$, and let $z$ be a collision-free local maximizer of $\Delta_n$ on $\mathcal{S}_n$. Set*

$$t_i = |z_i|^2, \quad s = \sum_i (t_i - 1)^2, \quad p_2 = \sum_i z_i^2, \quad \lambda = \frac{n-1}{2}.$$

*Then*

$$\sum_{j \ne i} \frac{1}{z_i - z_j} = \lambda \overline{z_i}, \qquad \sum_i z_i = 0. \tag{2.1}$$

*The complex symmetric matrix*

$$B_{ij} = -\frac{1}{(z_i - z_j)^2} \quad (i \ne j), \qquad B_{ii} = \sum_{j \ne i} \frac{1}{(z_i - z_j)^2} \tag{2.2}$$

*satisfies*

$$B\mathbf{1} = 0, \qquad Bz = \lambda \overline{z}, \qquad \|B\|_{\mathrm{op}} \le \lambda. \tag{2.3}$$

*Moreover,*

$$3ns + |p_2|^2 \le n^2, \qquad |z_i - z_j|^2 \ge \frac{2n}{(n-1)^2}. \tag{2.4}$$

*If $n \ge 3$, the second power sum also satisfies*

$$|p_2|^2 \le \frac{ns(n+s)}{s + n(n-2)^2/(n-1)^2}. \tag{2.5}$$

*For the real weighted Laplacian with weights $|z_i - z_j|^{-2}$, one has*

$$Lt = (n-1)(t - \mathbf{1}). \tag{2.6}$$

*Proof.* Lagrange multipliers for $\log \Delta_n$ give the first equation in (2.1) with a real multiplier. Multiplying by $z_i$ and summing yields $\lambda n = n(n-1)/2$; summing the equations gives the zero centroid. The first two identities in (2.3) follow directly.

For $\langle z, h\rangle = 0$, every phase multiple of $h$ is a real tangent vector to $\mathcal{S}_n$. The second derivative of the Lagrangian is

$$-2\,\mathrm{Re}(h^{\mathsf{T}} B h) - 2\lambda \|h\|^2.$$

Its nonpositivity for every phase implies $|h^{\mathsf{T}}Bh| \le \lambda\|h\|^2$. Write any vector as $h+\alpha z$ with $\langle z, h\rangle = 0$. Since $h^{\mathsf{T}}Bz = 0$ and $z^{\mathsf{T}}Bz = \lambda n$, the same quadratic bound holds for all vectors. For unit vectors $x, y$, the identity

$$4x^{\mathsf{T}}By = (x+y)^{\mathsf{T}}B(x+y) - (x-y)^{\mathsf{T}}B(x-y)$$

and the parallelogram law give $|x^{\mathsf{T}}By| \le \lambda$. Taking the supremum proves the operator norm bound.

Pairwise expansion gives $Bz^{\circ 2} = (n-1)(t-\mathbf{1})$. Applying (2.3) to $z^{\circ 2} - p_2\mathbf{1}/n$ yields

$$(n-1)^2 s \le \lambda^2 \left( \sum_i t_i^2 - \frac{|p_2|^2}{n} \right).$$

As $\sum_i t_i^2 = n + s$, this is the first bound in (2.4). To sharpen the entry bound, let $P_0 = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/n$. Since $B = P_0 B P_0$ and $\|P_0 e_i\|^2 = (n-1)/n$, we have $|B_{ij}| \le \lambda(n-1)/n$. This proves the second inequality. To prove (2.5), multiply the stationary equations by $z_i^3$ and sum. Since $\sum_i z_i = 0$, pairwise expansion gives $\sum_i t_i z_i^2 = (2n-3)p_2/(n-1)$. Thus

$$\langle t - \mathbf{1}, z^{\circ 2} - p_2\mathbf{1}/n \rangle = \frac{n-2}{n-1} p_2.$$

Cauchy–Schwarz yields $((n-2)/(n-1))^2 |p_2|^2 \le s(n+s-|p_2|^2/n)$. Rearranging proves (2.5), including $s = 0$. Finally, multiply (2.1) by $z_i$, take real parts, and use $2\,\mathrm{Re}(z_i/(z_i - z_j)) = 1 + (t_i - t_j)/|z_i - z_j|^2$. This gives (2.6). □

### 2.2 Five-point moments and a Gram matrix

From now on $n = 5$, and we work at a collision-free local maximizer. In addition to $t_i$ and $s$, put

$$\begin{aligned} T &= \sum_i t_i^2 = 5 + s, \qquad U = \sum_i t_i^3, \qquad V = \sum_i t_i^4, \\ p_j &= \sum_i z_i^j, \quad q = |p_2|^2, \quad p = |p_3|^2, \quad w = |p_4|, \quad m = \sum_i t_i^2 z_i^2. \end{aligned} \tag{2.7}$$

Proposition 2.1 gives

$$0 \le s \le \frac{5}{3}, \qquad 0 \le q \le \min\left\{ 25 - 15s, \frac{80s(5+s)}{45+16s} \right\}. \tag{2.8}$$

Multiplication of the stationary equations by $z_i^2, z_i^3, z_i^4$, followed by summation and pairing, gives

$$\sum_i t_i z_i = 0, \qquad \sum_i t_i z_i^2 = \frac{7}{4} p_2, \qquad \sum_i t_i z_i^3 = \frac{3}{2} p_3. \tag{2.9}$$

For example, for the middle identity the pair sum is $4p_2 + \sum_{i<j} z_i z_j = 7p_2/2$. Direct expansion also gives

$$Bz^{\circ 2} = 4(t - \mathbf{1}), \qquad Bz^{\circ 4} = (8t - 9\mathbf{1}) \circ z^{\circ 2} - p_2\mathbf{1}. \tag{2.10}$$

The matrix $4I - B^*B$ is positive semidefinite. Let $D$ be its Gram matrix on the two vectors $z^{\circ 2} - p_2\mathbf{1}/5$ and $z^{\circ 4} - p_4\mathbf{1}/5$. Using (2.9)–(2.10), its entries are

$$\begin{aligned} D_{22} &= 20 - 12s - \frac{4}{5} q, \\ D_{44} &= -60V + 144U - 81T + 5q - \frac{4}{5} w^2, \\ D_{24} &= -28m + 83p_2 - \frac{4}{5} \overline{p_2} p_4. \end{aligned} \tag{2.11}$$

In particular,

$$D_{22}, D_{44} \ge 0, \qquad |D_{24}|^2 \le D_{22} D_{44}. \tag{2.12}$$

## 3 A two-dimensional projection constraint

**Lemma 3.1.** *For $s > 0$ at a collision-free local maximizer,*

$$w \geq T - \frac{2q}{5} - \frac{9q}{8s} - \frac{p}{5-\sqrt{q}}. \tag{3.1}$$

*Proof.* By (2.8), $q < 25$. Thus $E = \operatorname{span}_{\mathbb{C}}\{\mathbf{1}, z, \overline{z}\}$ has dimension three: $\mathbf{1}$ is orthogonal to the other two vectors, whose Gram determinant is $25 - q > 0$. The vector $u = t - \mathbf{1}$ is real, lies in $E^{\perp}$, and has norm squared $s$. Let $v$ be the orthogonal projection of $z^{\circ 2}$ onto $E^{\perp}$. The Gram matrix of $z, \overline{z}$ is $\left(\begin{smallmatrix} 5 & \overline{p_2} \\ p_2 & 5 \end{smallmatrix}\right)$. Since $\langle z, z^{\circ 2}\rangle = 0$ and $\langle \overline{z}, z^{\circ 2}\rangle = p_3$, the linear part of the projection onto $E$ is

$$\frac{-\overline{p_2} p_3 z + 5 p_3 \overline{z}}{25 - q}.$$

Consequently,

$$\begin{aligned} \|v\|^2 &= T - \frac{q}{5} - \frac{5p}{25-q}, \qquad \langle u, v\rangle = \frac{3}{4} p_2, \\ v^{\mathsf{T}} v &= p_4 - \frac{p_2^2}{5} + \frac{\overline{p_2} p_3^2}{25-q}. \end{aligned} \tag{3.2}$$

The conjugation-invariant space $E^{\perp}$ has a real orthonormal basis $u/\sqrt{s}, e$. Write $v = a\, u/\sqrt{s} + be$. Then $|a|^2 = 9q/(16s)$ and

$$|v^{\mathsf{T}} v| = |a^2 + b^2| \geq \|v\|^2 - 2|a|^2 = \|v\|^2 - \frac{9q}{8s}.$$

Combine this with (3.2) and the triangle inequality. The coefficient of $p$ becomes $(5+\sqrt{q})/(25-q) = 1/(5-\sqrt{q})$, proving the assertion. □

## 4 Excluding small nonzero radial variance

**Proposition 4.1** (Local radial gap)**.** *A collision-free local maximizer satisfies either $s = 0$ or $s > 1/20$.*

*Proof.* Suppose $0 < s \leq 1/20$, and write $z_i = r_i e^{i\theta_i}$. For $\delta_i = t_i - 1$, we have $\sum_i \delta_i = 0$ and $\sum_i \delta_i^2 = s$. Hence

$$t_i \geq 1 - \sqrt{4s/5} \geq \frac{4}{5}, \qquad \frac{7}{8} < r_i r_j < \frac{9}{8}. \tag{4.1}$$

For completeness, set $a = (t_i + t_j)/2$, $b = (t_i - t_j)/2$. Cauchy–Schwarz for the remaining three deviations gives $s \geq (10/3)(a-1)^2 + 2b^2$. It follows that $t_i t_j \geq a^2 - s/2 + (5/3)(a-1)^2$. This last expression is increasing for $a \geq 1 - \sqrt{3s/10} > 7/8$, since its derivative is $(16a - 10)/3$. At the left endpoint it equals $a^2$, so $r_i r_j \geq 1 - \sqrt{3s/10} > 7/8$. The upper bound follows from $r_i r_j \leq a \leq 1 + \sqrt{3s/10} < 9/8$; here $(1/8)^2 > 3/200$.

Let $\overline{r} = 5^{-1}\sum_i r_i$ and $\sigma = \sum_i (r_i - \overline{r})^2$. Using $(r_i + r_j)^2 \geq 4 r_i r_j > 7/2$,

$$\sigma = \frac{1}{5}\sum_{i<j} \frac{(t_i - t_j)^2}{(r_i + r_j)^2} < \frac{2s}{7} \leq \frac{1}{70}, \qquad (r_i - r_j)^2 < \frac{4s}{7} \leq \frac{1}{35}. \tag{4.2}$$

All squared distances are at least $1/2$ by Proposition 2.1. Thus

$$\sin^2 \frac{\theta_i - \theta_j}{2} = \frac{|z_i - z_j|^2 - (r_i - r_j)^2}{4 r_i r_j} > \frac{11}{105} > \frac{1}{10}.$$

The angles are distinct and all the corresponding half-angle cotangents have absolute value less than 3.

Lift and order the angles so that $\theta_1 < \cdots < \theta_5 < \theta_1 + 2\pi$, and put

$$\mathcal{C}(\theta) = \sum_{i<j} \log |e^{i\theta_i} - e^{i\theta_j}|^2.$$

For $\sum_i h_i = 0$,

$$h^{\mathsf{T}} \nabla^2 \mathcal{C}(\theta) h = -\frac{1}{2} \sum_{i<j} \csc^2 \frac{\theta_i - \theta_j}{2} (h_i - h_j)^2 \le -\frac{5}{2} \|h\|^2.$$

Let $\theta^*$ be the equally spaced angles with the same mean, and let $\eta = \|\theta - \theta^*\|$. The ordered region of span less than $2\pi$, intersected with the fixed-mean hyperplane, is convex. Integration of the preceding Hessian bound along the connecting segment gives

$$\eta \le \frac{2}{5} \|\nabla \mathcal{C}(\theta)\|. \tag{4.3}$$

The angular stationary equations for the actual radii imply

$$\frac{\partial \mathcal{C}}{\partial \theta_i} = \sum_{j \ne i} \cot \frac{\theta_i - \theta_j}{2} \frac{(r_i - r_j)^2}{|z_i - z_j|^2}. \tag{4.4}$$

The incidence matrix of $K_5$ has operator norm $\sqrt{5}$. For $v_i = r_i - \overline{r}$, the zero sum gives $\max_i v_i^2 \le 4\sigma/5$, and expansion yields $\sum_{i<j} (v_i - v_j)^4 = 5 \sum_i v_i^4 + 3\sigma^2 \le 7\sigma^2$. Combining these observations with (4.3)–(4.4), if squared distances are at least $d$ and cotangents at most $h$ in absolute value, then

$$\eta \le \gamma \sigma h / d, \qquad \gamma = \frac{2\sqrt{35}}{5}. \tag{4.5}$$

Initially $d = 1/2, h = 3$ and (4.2) give $\eta < 3\gamma/35 < 21/100$.

We next use only algebraic constants. Put

$$S_1 = \sqrt{\frac{5 - \sqrt{5}}{8}}, \quad C_1 = \frac{1 + \sqrt{5}}{4}, \qquad S_2 = \sqrt{\frac{5 + \sqrt{5}}{8}}, \quad C_2 = \frac{\sqrt{5} - 1}{4}, \tag{4.6}$$
$$P_j(u) = S_j(1 - u^2/4) - C_j u/\sqrt{2}, \qquad N(u) = C_1 + S_1 u/\sqrt{2}.$$

Here $S_j = \sin(j\pi/5)$, $C_j = \cos(j\pi/5)$. The inequalities $\cos x \ge 1 - x^2/2$ and $\sin x \le x$ for $x \ge 0$ give

$$\sin(j\pi/5 - u/\sqrt{2}) \ge P_j(u), \qquad \cos(\pi/5 - u/\sqrt{2}) \le N(u).$$

If $\eta \le u \le 21/100$, each pairwise angular error is at most $\sqrt{2}u$. Consequently every absolute half-angle sine is at least $P_1(u) > 0$ and the absolute cotangent is at most $N(u)/P_1(u)$. Equations (4.1), (4.2), and (4.5) then give

$$\eta < \frac{\gamma N(u)}{245 P_1(u)^3}. \tag{4.7}$$

The exact algebraic checks in the ancillary program prove

$$\frac{\gamma N(21/100)}{245 P_1(21/100)^3} < \frac{9}{100}, \qquad \frac{\gamma N(9/100)}{245 P_1(9/100)^3} < \frac{3}{50}. \tag{4.8}$$

Thus $\eta < 3/50$. The same algebraic checks give

$$\frac{2}{7P_1(3/50)^2} < \frac{19}{20}, \qquad \frac{2}{7P_2(3/50)^2} < \frac{1}{3}. \tag{4.9}$$

These bound the inverse squared distances on consecutive pairs and diagonals in the cyclic order, respectively.

Let $L$ be the real Laplacian with weights $|z_i - z_j|^{-2}$. Comparison in the positive semidefinite order with a pentagon of edge weight $19/20$ and diagonal weight $1/3$ gives

$$\lambda_{\max}(L) \le \frac{19}{20}\frac{5+\sqrt{5}}{2} + \frac{1}{3}\frac{5-\sqrt{5}}{2} < 4. \tag{4.10}$$

The two Laplacians of the pentagon and its complement are simultaneously diagonalized by the discrete Fourier basis; their nonzero eigenvalues are $(5 \pm \sqrt{5})/2$ in complementary order. However, (2.6) gives $L(t - \mathbf{1}) = 4(t - \mathbf{1})$. As $s > 0$, this contradicts (4.10). □

## 5 A quartic logarithm majorant

Put

$$a = \frac{5-\sqrt{5}}{2}, \quad b = \frac{5+\sqrt{5}}{2}, \quad c = \frac{1}{5} - \frac{2}{5\sqrt{5}}\log\frac{b}{a}, \quad k = \frac{9}{2500}. \tag{5.1}$$

The following bounds are certified in the ancillary program:

$$\frac{22}{3}k < c < C := \frac{278365}{10000000} < \frac{17}{2}k. \tag{5.2}$$

Let $H$ be the cubic interpolating $\log x$ and its derivative at $a, b$. Solving the four interpolation conditions gives

$$H(x) = \frac{1}{2}\log 5 - \frac{7}{4} - \frac{25c}{4} + (1+15c)x - \left(\frac{1}{10} + \frac{15c}{2}\right)x^2 + cx^3. \tag{5.3}$$

**Lemma 5.1.** *For $0 < x \le R := 41/6$,*

$$\log x \le H(x) - k(x^2 - 5x + 5)^2. \tag{5.4}$$

*Proof.* Add the node $R$ to the double nodes $a, b$. The quartic Hermite interpolant is $H(x) - k_R(x^2 - 5x + 5)^2$, where

$$k_R = \frac{H(R) - \log R}{(R^2 - 5R + 5)^2}.$$

The fifth derivative of $\log x$ is positive. Its Hermite remainder has the sign of $(x-a)^2(x-b)^2(x-R)$, hence is nonpositive for $0 < x \le R$. The exact series check in Appendix A proves $k_R > k$. This yields (5.4). □

All squared distances at a local maximizer belong to this range:

$$|z_i - z_j|^2 \le 2(t_i + t_j) \le 4 + 2\sqrt{6s/5} \le 4 + 2\sqrt{2} < 41/6.$$

Write $d_{ij} = |z_i - z_j|^2$ and $\mathcal{Q} = \sum_{i<j}(d_{ij}^2 - 5d_{ij} + 5)^2$. Expansion using (2.9) gives

$$\begin{aligned}
\sum_{i<j} d_{ij} &= 25, \\
\sum_{i<j} d_{ij}^2 &= 5T + 50 + q, \\
\sum_{i<j} d_{ij}^3 &= 5U + 45T + \frac{21}{2}q - p, \\
\sum_{i<j} d_{ij}^4 &= 5V + 80U + 18T^2 + 49q - 12p + w^2 + 12\,\mathrm{Re}(m\overline{p_2}).
\end{aligned} \tag{5.5}$$

These identities can be checked uniformly by putting $M_{\alpha\beta} = \sum_i z_i^\alpha \overline{z_i}^\beta$ and using

$$\sum_{i<j} d_{ij}^r = \frac{1}{2}\sum_{\alpha,\beta=0}^{r} (-1)^{\alpha+\beta}\binom{r}{\alpha}\binom{r}{\beta} M_{r-\alpha,r-\beta}M_{\alpha\beta}. \tag{5.6}$$

The regular pentagon has five squared distances $a$ and five $b$. Subtract its value in (5.4) and use (5.5) to obtain

$$\log\Delta_5 - 5\log 5 \le \left(-\frac{1}{2} + \frac{15c}{2}\right)s + 5c(U-5) + \left(-\frac{1}{10} + 3c\right)q - cp - k\mathcal{Q}. \tag{5.7}$$

The moment expansion of $\mathcal{Q}$ is

$$\begin{aligned}
\mathcal{Q} = {} & 5V + 30U + 18T^2 - 275T + 750 - 21q - 2p + w^2 \\
& + 12\,\mathrm{Re}(m\overline{p_2}).
\end{aligned} \tag{5.8}$$

Set

$$R_0 = 5V + 30U + 18T^2 - 275T + 750, \qquad D_0 = -60V + 144U - 81T + 5q.$$

Solving (2.11) for $m$ and applying (2.12) gives

$$\mathcal{Q} \ge R_0 + \frac{102}{7}q - 2p + w^2 - \frac{12}{35}qw - \frac{3}{7}\sqrt{qD_{22}D_0}. \tag{5.9}$$

Indeed, the contribution of $m$ is

$$12\,\mathrm{Re}(m\overline{p_2}) = \frac{249}{7}q - \frac{12}{35}\,\mathrm{Re}(\overline{p_2}^2 p_4) - \frac{3}{7}\,\mathrm{Re}(D_{24}\overline{p_2}).$$

Also $D_0 = D_{44} + 4w^2/5 \ge D_{44} \ge 0$, so replacing $D_{44}$ by $D_0$ in the square root weakens the lower bound in the required direction.

# 6 Reduction to two variables

## 6.1 Eliminating the complex power sums

Assume $s > 0$. Put $L_0 = T - 2q/5 - 9q/(8s)$ and $w_0 = 6q/35$. Lemma 3.1 implies $p \ge (5 - \sqrt{q})[L_0 - w]_+$. The expression to be bounded below after combining (5.7) and (5.9) is $k(w^2 - 12qw/35) + (c - 2k)p$. It satisfies

$$k(w^2 - 12qw/35) + (c-2k)p \ge k\left([T - \left(\frac{4}{7} + \frac{9}{8s}\right)q]_+^2 - \frac{36q^2}{1225}\right). \tag{6.1}$$

To see this, complete the square at $w_0$. If $L_0 \le w_0$, the unconstrained minimum at $w_0$ is feasible. If $L_0 > w_0$, the minimum is at $L_0$ provided

$$L_0 - w_0 \le \frac{c-2k}{2k}(5-\sqrt{q}).$$

Here $L_0 > 0$ implies $q < T/(2/5 + 9/(8s)) \le 800/129 < 25/4$. By (5.2), the last right-hand side is greater than $(8/3)(5/2) = 20/3 \ge T \ge L_0 - w_0$. This proves the asserted piecewise minimum.

### 6.2 Eliminating the radial moments

Cauchy–Schwarz and projection of $t^{\circ 2}$ onto the orthogonal vectors $\mathbf{1}, t - \mathbf{1}$ give

$$U \ge \frac{T^2}{5}, \qquad V \ge V_0(U,s) := \frac{T^2}{5} + \frac{(U-T)^2}{s}. \tag{6.2}$$

The zero-sum deviations $\delta_i = t_i - 1$ also satisfy

$$\sum_i \delta_i^3 \ge -\frac{3}{\sqrt{20}} s^{3/2}.$$

For a proof, at an extremum of the third moment on $\sum \delta_i = 0$, $\sum \delta_i^2 = s$, the multiplier equation is a quadratic in $\delta_i$. Thus at most two values occur; checking multiplicities $1+4$ and $2+3$ gives the stated sharp lower bound. Consequently,

$$U \ge U_0(s) = \begin{cases} 5 + 3s - \dfrac{3}{\sqrt{20}} s^{3/2}, & 0 < s \le 5/4, \\ (5+s)^2/5, & 5/4 \le s \le 5/3. \end{cases} \tag{6.3}$$

The two candidates meet at $s = 5/4$, as follows by setting $x = \sqrt{s/20}$ and factoring $1 - 3x - 4x^2 = (1-4x)(1+x)$.

After (6.1), the right-hand side of the logarithm bound decreases with $V$: its nonradical derivative is $-5k$ and the radicand decreases with $V$. We may therefore set $V = V_0(U,s)$. For $U \ge U_0(s)$,

$$\partial_U V_0 = \frac{2(U-T)}{s} \ge \begin{cases} 4 - 3\sqrt{s/5} \ge 5/2, & s \le 5/4, \\ 2(5+s)/5 \ge 5/2, & s \ge 5/4. \end{cases}$$

Hence $\partial_U D_0 \le -6$, while the remaining derivative is at most $5c - (85/2)k < 0$. We may also set $U = U_0(s)$. Both downward substitutions increase $D_0$, preserving its nonnegativity along the substitution path. Finally the resulting upper bound increases with $c$, with derivative $(15/2)s + 5(U_0 - 5) + 3q > 0$, so we may replace $c$ by the rational upper bound $C$ in (5.2).

Define

$$\begin{aligned} U_* = U_0(s), \quad V_* &= \frac{(5+s)^2}{5} + \frac{(U_* - 5 - s)^2}{s}, \\ A_q = \frac{1}{10} - 3C + \frac{102k}{7}, \quad B_q &= \frac{36k}{1225}, \quad D_q = \frac{3k}{7}, \\ f(s) &= \left(-\frac{1}{2} + \frac{15C}{2}\right) s + 5C(U_* - 5) \\ &\quad -k\{5V_* + 30U_* + 18(5+s)^2 - 275(5+s) + 750\}, \\ D_*(s) &= -60V_* + 144U_* - 81(5+s). \end{aligned} \tag{6.4}$$

We have proved the following scalar reduction.

**Proposition 6.1.** *At a collision-free local maximizer with $s > 0$,*

$$\log \Delta_5 - 5 \log 5 \le F(s,q), \tag{6.5}$$

*where*

$$\begin{aligned} F(s,q) = f(s) &- A_q q + B_q q^2 \\ &+ D_q \sqrt{q(20 - 12s - 4q/5)(D_*(s) + 5q)} \\ &- k[5 + s - \left(\frac{4}{7} + \frac{9}{8s}\right) q]_+^2. \end{aligned} \tag{6.6}$$

*The parameters satisfy (2.8), and both nontrivial factors inside the square root are nonnegative.*

# 7 Exact certificate and completion of the proof

**Lemma 7.1** (Scalar certificate)**.** *With the constants in (6.4),*

$$F(s,q) < 0 \qquad \left(\frac{1}{20} \le s \le \frac{5}{3}, \quad 0 \le q \le \min\left\{25 - 15s, \frac{80s(5+s)}{45+16s}\right\}\right). \tag{7.1}$$

*Proof.* We treat (7.1) in two regions:

$$\mathrm{A}: \ s = 20x^2, \quad \frac{1}{20} \le x \le \frac{1}{4}; \qquad \mathrm{B}: \ \frac{5}{4} \le s \le \frac{5}{3}.$$

In A, $U_* = 5 + 60x^2 - 60x^3$ and $V_* = 5 + 120x^2 - 240x^3 + 260x^4$. In B, $U_* = (5+s)^2/5$ and $V_* = (5+s)^3/25$. The endpoint $20(1/20)^2 = 1/20$ exactly matches the local gap, and the regions meet at $s = 5/4$.

Region A starts with its containing rectangle and $0 \le q \le 25$. For a rectangle $[x_l, x_h] \times [q_l, q_h]$, put $s_{\min} = 20x_l^2$, $s_{\max} = 20x_h^2$, and replace $q_h$ by

$$\min\left\{q_h, 25 - 15s_{\min}, \frac{80s_{\max}(5+s_{\max})}{45+16s_{\max}}\right\}.$$

The last expression is justified by (2.5) and the monotonicity of $80s(5+s)/(45+16s)$ for $s \ge 0$. Discard an empty rectangle. Evaluate the interval form of (6.6), clipping its radicand factors to their nonnegative parts. If the upper endpoint is negative, accept the rectangle. Otherwise bisect its longer normalized side, using scales $1/5$ and $1$. The depth-first search exhausts all rectangles.

| Region | Visited nodes | Accepted rectangles | Maximum depth |
|---|---|---|---|
| A | 183 | 92 | 21 |

No rectangles remain unresolved. Every accepted upper endpoint is less than the following convenient rational bound:

$$-\frac{1}{200000}. \tag{7.2}$$

All comparisons use exact fractions. The interval arithmetic and the constant checks are detailed in Appendix A. Pruning removes only infeasible points and each bisection covers its parent, so the finite certified cover proves the assertion.

On the second branch no subdivision is necessary. In fact,

$$F(s,q) \le -\frac{56657645489}{1713169664000} < -\frac{3}{100} \qquad \left(\frac{5}{4} \le s \le \frac{5}{3}, \quad 0 \le q \le 25 - 15s\right). \tag{7.3}$$

To see this, write

$$f(s) = a_0 s + b_0 s^2 - \frac{k}{5}s^3, \quad a_0 = -\frac{1}{2} + \frac{35}{2}C + 20k, \quad b_0 = C - 27k.$$

The rational inequalities $b_0 < 0$ and $a_0 + (5/2)b_0 < 0$ imply that $f$ is decreasing for $s \geq 5/4$. Likewise,

$$D_*(s) = 15 + 27s - \frac{36}{5}s^2 - \frac{12}{5}s^3$$

is decreasing on this interval, since its derivative is at most $-9/4$. Consequently $f(s) \leq -9181/256000$, $D_*(s) \leq 525/16$, $q \leq 25/4$, $20 - 12s - (4/5)q \leq 5$, and $D_*(s) + 5q \leq 1025/16$. Put

$$\alpha = A_q - \frac{25}{4}B_q = \frac{6692069}{98000000} > 0.$$

Dropping the nonpositive cutoff term in $F$ and completing the square in $\sqrt{q}$ gives

$$F(s,q) \leq -\frac{9181}{256000} - \alpha q + D_q\sqrt{\frac{5125}{16}q} \leq -\frac{9181}{256000} + \frac{5125D_q^2}{64\alpha} = -\frac{56657645489}{1713169664000}.$$

Every constant comparison in this argument is checked exactly in the ancillary program. □

*Proof of Theorem 1.1.* The constraint sphere is compact, and a regular pentagon gives $\Delta_5 = 5^5 > 0$. A global maximizer exists and is collision-free. If $0 < s \leq 1/20$, Proposition 4.1 excludes it. If $s \geq 1/20$, Proposition 6.1 and Lemma 7.1 give $\Delta_5 < 5^5$, a contradiction. Thus $s = 0$ at every global maximizer, so $|z_i| = 1$.

For completeness, the classical unit-circle conclusion can be read directly from the root polynomial, as in [1]. Let $P(\zeta) = \prod_i(\zeta - z_i)$. The stationary equations give

$$\frac{P''(z_i)}{2P'(z_i)} = \sum_{j\neq i}\frac{1}{z_i - z_j} = 2\overline{z_i} = \frac{2}{z_i}.$$

The polynomial $\zeta P''(\zeta) - 4P'(\zeta)$ has degree at most three and vanishes at five distinct points, so it is zero. Coefficient comparison gives $P(\zeta) = \zeta^5 - \alpha$, $|\alpha| = 1$. The roots are precisely a unit regular pentagon, and $\Delta_5 = \prod_i |P'(z_i)| = 5^5$. This establishes both the inequality and all equality cases. □

*Remark* 7.2. The local gap is a statement about every local maximizer, whereas (6.5) need not exclude a local maximizer whose value is strictly below $5^5$. The global comparison in the final proof uses the known regular-pentagon competitor.

## 8 A general moment hierarchy and its limitations

The second-order method extends to arbitrary $n$, although the target $\Delta_n \leq n^n$ does not: Chalk's counterexample already disproves this bound for every $n \geq 6$ [2]. We record constraints that remain valid at all collision-free local maximizers and a quantitative six-point consequence. The polygonal spectral calculation below records a limitation of the method; it is not presented as a new stability result.

### 8.1 Higher-degree second-order constraints

Work at a collision-free local maximizer on $\mathcal{S}_n$, and put $p_k = \sum_i z_i^k$, $p_0 = n$, and $\lambda = (n-1)/2$. For every $k \geq 1$, the stationary equations imply

$$\sum_i t_i z_i^k = \frac{(2n-k-1)p_k + \sum_{j=1}^{k-1} p_j p_{k-j}}{n-1}. \tag{8.1}$$

Indeed, multiply the stationary equation by $z_i^{k+1}$, sum, and pair indices; the result before division by $\lambda$ is $\frac{1}{2}\{\sum_{j=0}^{k} p_j p_{k-j} - (k+1)p_k\}$. For $k \geq 2$ a related coordinatewise identity is

$$Bz^{\circ k} = (k\lambda t - (k-1)(n-k/2)\mathbf{1}) \circ z^{\circ(k-2)} - \sum_{j=1}^{k-2}(k-1-j)p_j z^{\circ(k-2-j)}. \tag{8.2}$$

To verify it, expand $z_i^k - z_j^k = (z_i - z_j)\sum_{r=0}^{k-1} z_i^{k-1-r} z_j^r$ and then use, for $r \geq 1$,

$$\sum_{j \neq i} \frac{z_j^r}{z_i - z_j} = \lambda z_i^r \overline{z}_i - \sum_{\ell=0}^{r-1} p_\ell z_i^{r-1-\ell} + r z_i^{r-1}.$$

**Proposition 8.1** (Moment Gram hierarchy)**.** *For any finite set of positive integers $\mathcal{K}$, set $x_k = z^{\circ k} - p_k\mathbf{1}/n$ and define*

$$D_{k\ell} = \lambda^2 \langle x_k, x_\ell \rangle - \langle Bz^{\circ k}, Bz^{\circ \ell} \rangle, \qquad k, \ell \in \mathcal{K}. \tag{8.3}$$

*Then $D$ is Hermitian positive semidefinite. Its entries are polynomials in mixed moments, with the identities (8.1)–(8.2) providing exact reductions.*

*Proof.* The matrix $\lambda^2 I - B^*B$ is positive semidefinite by Proposition 2.1, and $B\mathbf{1} = 0$. Thus $D$ is its Gram matrix on the vectors $x_k$. Constant projection removes a nonnegative term that would otherwise weaken the constraints. □

For an explicit reusable instance, write

$$T = \sum_i t_i^2 = n + s, \quad U = \sum_i t_i^3, \quad V = \sum_i t_i^4, \quad q = |p_2|^2, \quad w = |p_4|, \quad m = \sum_i t_i^2 z_i^2.$$

The submatrix on $\mathcal{K} = \{2, 4\}$ has entries

$$\begin{aligned} D_{22} &= \frac{(n-1)^2}{4}\left(n - 3s - \frac{q}{n}\right), \\ D_{44} &= -\frac{15(n-1)^2}{4}V + 12(n-1)(n-2)U - 9(n-2)^2 T + nq - \frac{(n-1)^2}{4n}w^2, \\ D_{24} &= -\frac{7(n-1)^2}{4}m + (7n^2 - 22n + 18)p_2 - \frac{(n-1)^2}{4n}\overline{p_2}p_4. \end{aligned} \tag{8.4}$$

They satisfy $D_{22}, D_{44} \geq 0$ and $|D_{24}|^2 \leq D_{22}D_{44}$. These formulas follow from $Bz^{\circ 4} = (2(n-1)t - 3(n-2)\mathbf{1}) \circ z^{\circ 2} - p_2\mathbf{1}$ and (8.1), and reduce exactly to (2.11) when $n = 5$. Increasing $\mathcal{K}$ supplies additional necessary constraints; it does not by itself prove a global upper bound.

### 8.2 The dimensional obstruction in the projection step

Assume $n \geq 3$ and $s > 0$, set $p = |p_3|^2$, and let $E = \operatorname{span}_{\mathbb{C}}\{\mathbf{1}, z, \overline{z}\}$. The radial bound gives $q < n^2$, so $\dim E = 3$. Let $u = t - \mathbf{1}$ and let $v$ be the projection of $z^{\circ 2}$ onto $E^{\perp}$. The computation in Lemma 3.1 now gives

$$\begin{aligned} \|v\|^2 &= T - \frac{q}{n} - \frac{np}{n^2 - q}, \qquad \langle u, v \rangle = \frac{n-2}{n-1} p_2, \\ v^{\mathsf{T}} v &= p_4 - \frac{p_2^2}{n} + \frac{\overline{p_2} p_3^2}{n^2 - q}. \end{aligned} \tag{8.5}$$

Since $u \in E^{\perp}$ and $\|u\|^2 = s$, Cauchy–Schwarz proves the dimension-independent bound

$$\frac{(n-2)^2}{(n-1)^2} q \leq s \left( T - \frac{q}{n} - \frac{np}{n^2 - q} \right). \tag{8.6}$$

Dropping the last nonpositive term recovers (2.5). For $n = 5$, $E^{\perp}$ admits an orthonormal basis of two real vectors, and the further bound on $|v^{\mathsf{T}} v|$ used earlier is available. For $n \geq 6$ that particular argument fails: after fixing the real direction $u$, two real orthonormal directions $e_1, e_2$ remain, and $b(e_1 + ie_2)$ has positive Hermitian norm but zero transpose square. Thus no analogous lower bound follows solely from the norm and the component along $u$. Additional moment constraints would be needed.

### 8.3 A quantitative six-point radial gap

**Proposition 8.2.** *At a collision-free local maximizer of $\Delta_6$ on $\mathcal{S}_6$, either $s = 0$ or $s > 1/200$.*

*Proof.* Suppose $0 < s \leq 1/200$, and write $z_i = r_i e^{i\theta_i}$. For $a = (t_i + t_j)/2$ and $b = (t_i - t_j)/2$, the zero-sum deviations give $s \geq 3(a-1)^2 + 2b^2$. The function $a^2 - s/2 + \frac{3}{2}(a-1)^2$ is increasing for $a > 3/5$, so its value at $a = 1 - \sqrt{s/3}$ proves

$$\frac{23}{24} < 1 - \sqrt{s/3} \leq r_i r_j \leq 1 + \sqrt{s/3} < \frac{25}{24}.$$

Here $s/3 \leq 1/600 < 1/576$. With $\sigma = \sum_i (r_i - \overline{r})^2$, the same pairwise variance identities as in the five-point proof give

$$\sigma = \frac{1}{6} \sum_{i<j} \frac{(t_i - t_j)^2}{(r_i + r_j)^2} < \frac{6s}{23}, \qquad (r_i - r_j)^2 < \frac{12s}{23} \leq \frac{3}{1150}.$$

The strengthened general distance bound gives $d_{ij} \geq 12/25$. Consequently,

$$\sin^2 \frac{\theta_i - \theta_j}{2} > \frac{12/25 - 3/1150}{25/6} > \frac{1}{10}; \qquad \left| \cot \frac{\theta_i - \theta_j}{2} \right| < 3.$$

Order the lifted angles and let $\theta^*$ be the equally spaced six-point angles with the same mean. The circle logarithmic product has Hessian at most $-3I$ on the mean-zero subspace. Its gradient at the actual angles is given by (4.4), with six indices in place of five. The incidence matrix of $K_6$ has norm $\sqrt{6}$, while, for $v_i = r_i - \overline{r}$,

$$\sum_{i<j} (v_i - v_j)^4 = 6 \sum_i v_i^4 + 3\sigma^2 \leq 8\sigma^2.$$

Thus the angular error satisfies

$$\eta := \|\theta - \theta^*\| \le \frac{4}{\sqrt{3}}\frac{\sigma h}{d} < \frac{7}{3}\frac{6s}{23}\,3\,\frac{25}{12} = \frac{175s}{46} \le \frac{7}{368} < \frac{1}{50}.$$

Every half-angle error is at most $\eta/\sqrt{2} < 1/70$, since $\sqrt{2} > 7/5$. As sine is 1-Lipschitz, the absolute half-angle sines in the three cyclic separation classes are greater than $17/35$, $59/70$, and $69/70$, respectively; here $\sqrt{3}/2 > 6/7$. Multiplication by $4r_ir_j > 23/6$ gives

$$d_{ij} > \frac{9}{10}, \qquad d_{ij} > \frac{8}{3}, \qquad d_{ij} > \frac{100}{27}$$

in those three classes. Hence their Laplacian weights are less than $a_0 = 10/9$, $b_0 = 3/8$, $c_0 = 27/100$. The comparison circulant Laplacian has nonzero eigenvalues $a_0 + 3b_0 + 2c_0$, $3a_0 + 3b_0$, and $4a_0 + 2c_0$, so

$$\lambda_{\max}(L) \le \frac{2243}{450} < 5.$$

This contradicts $L(t - \mathbf{1}) = 5(t - \mathbf{1})$ and $s > 0$. All numerical comparisons are rational inequalities or follow by squaring positive rational numbers. □

### 8.4 Why a six-point global argument needs a different target

The inverse-distance Laplacian of the unit regular polygon similarly has eigenvalues $k(n-k)/2$, $0 \le k \le n-1$. Indeed, its $k$th eigenvalue is $\frac{1}{2}\sum_{r=1}^{n-1} |\sin(k\pi r/n)/\sin(\pi r/n)|^2$; expansion of $|\sum_{j=0}^{k-1} e^{2\pi ijr/n}|^2$ and Fourier orthogonality give $k(n-k)/2$. The largest eigenvalue is $\frac{1}{2}\lfloor n^2/4 \rfloor$. The strict spectral margin below $n-1$ holds for $2 \le n \le 6$, is zero at $n = 7$, and is absent for $n \ge 8$. Thus the local exclusion by strict Laplacian comparison cannot be copied unchanged for $n \ge 7$.

For completeness, Chalk's competitor consists of the origin and a regular $(n-1)$-gon of radius $\sqrt{n/(n-1)}$. It has

$$\Delta_n = (n-1)^{n-1}\left(\frac{n}{n-1}\right)^{n(n-1)/2}. \tag{8.7}$$

At $n = 6$ its ratio to $6^6$ is $6^9/5^{10} = 10077696/9765625 > 1$. For real $x \ge 6$, twice the logarithm of this ratio is $g(x) = x(x-3)\log x - (x-1)(x-2)\log(x-1)$, whose derivative is

$$g'(x) = (2x-3)\log\frac{x}{x-1} - 1 \ge \frac{2x-5}{2x-1} > 0.$$

The lower bound uses $\log(1+y) \ge 2y/(2+y)$ for $y \ge 0$, which follows by differentiation. Hence (8.7) exceeds $n^n$ for every integer $n \ge 6$. Proposition 8.2 therefore separates the regular hexagon from possible global maximizers, rather than proving that it is globally optimal. A full six-point sharp bound would require a new global comparison and additional constraints beyond the two-dimensional projection.

## A Exact arithmetic and reproducibility

### A.1 Interval operations

All endpoints in the program are instances of Python's `fractions.Fraction`. Addition, subtraction, multiplication, positive-part evaluation and squaring use their elementary outward interval formulas. Division is performed only when the denominator interval has positive lower endpoint.

For a nonnegative rational $v = a/b$, an outward square-root interval with denominator $2^{40}$ is obtained from

$$m = \operatorname{isqrt}\left(\left\lfloor a\,2^{80}/b \right\rfloor\right).$$

Its lower endpoint is $m/2^{40}$; the upper endpoint is this value when it squares exactly to $v$, and otherwise $(m+1)/2^{40}$. For a nondegenerate interval the lower and upper endpoint calculations are applied separately. Thus no floating-point approximation enters a mathematical decision. Wall-clock timings in the JSON output have no role in certification.

The factor $20 - 12s - 4q/5$ is nonnegative in the target domain. The other factor is also positive: in A,

$$D_* = 15 + x^2(-180 + 5760x - 15600x^2) \geq 15 - 180/16 > 0,$$

since $0 \leq x \leq 1/4$. In B, put $T = 5 + s \in [25/4, 20/3]$; then $D_* = T(-12T^2 + 144T - 405)/5 > 0$ by checking the endpoints of the concave quadratic. Clipping interval factors to their nonnegative parts is therefore valid, including rectangles extending outside the curved domain boundary.

### A.2 Transcendental and algebraic constants

The only transcendental checks concern logarithms. The coefficient $c$ has the positive-series representation

$$c = \frac{1}{5} - \frac{4}{25}\sum_{j=0}^{\infty} \frac{1}{(2j+1)5^j}.$$

After $N$ terms the series tail is at most $5/(4(2N+1)5^N)$. The program uses $N = 24$. For rational $x \geq 1$, with $y = (x-1)/(x+1)$,

$$\log x = 2\sum_{j=0}^{\infty} \frac{y^{2j+1}}{2j+1}.$$

After $N$ terms, its tail is at most $2y^{2N+1}/((2N+1)(1-y^2))$. Using $N = 120$ at $x = 5$ and $x = 41/6$ certifies (5.2) and $H(R) - \log R - k(R^2 - 5R + 5)^2 > 0$.

The checks in (4.8)–(4.10) use the algebraic constants (4.6), each enclosed by the integer-square-root procedure above. No numerical value of $\pi$ or trigonometric series is required. The underlying bounds $\sin x \leq x$ and $\cos x \geq 1 - x^2/2$ follow by integration of $\cos x \leq 1$ and $\sin x \leq x$ for $x \geq 0$.

### A.3 Polynomial expressions and source files

To make the interval encoding explicit, in A the functions in (6.4) simplify to

$$\begin{aligned} f(20x^2) &= x^2(-10 + 450C - 500k + x(-300C + 3000k - 8500kx)), \\ D_*(20x^2) &= 15 + x^2(-180 + 5760x - 15600x^2). \end{aligned} \tag{A.1}$$

In B, with $T = 5 + s$,

$$\begin{aligned} f(s) &= \left(-\frac{1}{2} + \frac{15C}{2}\right)s + 5C(T^2/5 - 5) - k(T^3/5 + 24T^2 - 275T + 750), \\ D_*(s) &= T(-12T^2 + 144T - 405)/5. \end{aligned} \tag{A.2}$$

The ancillary files supplied with this manuscript are:

- `verify_mordell.py`;
- `verification_result.json`;
- `algebra_identity_check.py`;
- `algebra_identity_result.json`;
- `generalization_check.py`;
- `generalization_result.json`.

From their directory, the full certificate is reproduced by

```
python verify_mordell.py --output verification_result.json
python algebra_identity_check.py
python generalization_check.py
```

The first command must be run with both regions, its default. The program exits unsuccessfully if any scalar region is unresolved or a constant comparison fails. Explicit exception checks remain active under Python's optimization option `-O`.

The second script independently expands (5.6) as formal polynomials over $\mathbb{Q}$, substitutes the mixed-moment relations, and checks all four identities in (5.5). It also derives (A.1)–(A.2) from (6.4) and checks exact point encodings. Its point checks are an encoding audit; the covering proof for region A of (7.1) is performed by the interval program, while region B is handled by (7.3). The third script checks the general Gram coefficients and supporting polynomial identities, the six-point rational constants, and the exact six-point value in (8.7). The recursive identities and the spectral and variational arguments are proved analytically above; finite script checks support those proofs and do not replace them.

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## References

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